

Predictive Temporal-Uncertainty Control (P-TUC) 2.0

Architectural Blueprint and Formal Proof of Structural Honesty

Building on the foundational contributions of Geoffrey Hinton, Yoshua Bengio, Yann LeCun, and Karl Friston

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1. Abstract

Building on concepts such as Forward-Forward, Equilibrium Prop, and Active Inference, we present **Predictive Temporal-Uncertainty Control (P-TUC)**, an architecture in which **coherent outputs emerge naturally from temporal and predictive constraints**.

By integrating a **Predictive Error-Gate** into a neuron's internal **Inference Clock**, silence becomes the default state for **uncertain or contradictory inputs**.

The evolution from **Temporal-Uncertainty Control (T-UC)** to P-TUC incorporates **phase-locking, predictive error gating, hierarchical validation, and local resonance plasticity**. The resulting architecture is **modular, scalable, energy-efficient, and compatible with neuromorphic hardware**, with a formal analysis demonstrating that structural mechanisms inhibit unreliable outputs.

Additionally, the design is **extensible**, supporting pluggable computation, gating, and learning modules.

2. Phase 1 — Original T-UC: Temporal Anchoring

Core Idea: Truth is related to **temporal stability**.

Mechanism:

- **Inference Clock (ω):** Signals propagate only after temporal coherence.
- **Signal Path:** Learned transformation of inputs.
- **Integration Window (T):** Outputs are realized only if the ~ 500 ms window shows stable inputs.

Ground State Realization:

$$y(t) = \begin{cases} f(Wx), & \text{if inputs stable over } T \\ 0, & \text{otherwise} \end{cases}$$

3. Phase 2 — Predictive Extension (P-TUC): The Honest Neuron

Core Idea: Replace static bias with **Prediction Error** (ϵ).

Mechanism:

Each neuron computes the **prediction error** by comparing the feedforward input S_{ff} with the top-down prediction P_{fb} .

Phase-Locked Loop (PLL) Mechanism:

- **Low Error:** Clock progresses $\rightarrow$ neuron ticks.
- **High Error:** Clock slows $\rightarrow$ neuron remains in Ground State.

Equations:

Local Error:

$$\epsilon(t) = |S_{\text{ff}}(t) - P_{\text{fb}}(t)|$$

Error-Gated Clock Evolution:

$$\frac{d\theta}{dt} = \omega_0 \cdot \text{sech}(\gamma\epsilon) + K \sum_i \sin(\psi_i - \theta)$$

Validated Tick:

$$y(t) = \begin{cases} f(Wx), & \text{if } \int_{t-T}^t \epsilon(\tau)^2 d\tau < \tau_{\text{threshold}} \\ 0, & \text{otherwise} \end{cases}$$

4. Phase 3 & 4 — Scaling, Open Architecture, and Extensibility

Socket-Based Design: Enables modular upgrades without changing Ground State behavior. Each socket provides a **pluggable interface**, allowing the architecture to scale and evolve.

Sockets Table:

Socket	Function	Equation / Logic	Notes
Comparator Socket (Error Logic)	Computes error	$\epsilon = S_{ff} - P_{fb} $	Pluggable: Bayesian Surprise, KL-Divergence, Contrastive Loss
Gating Socket (Predictive Brake)	Modulates clock	$\text{Brake} = \text{sech}(\gamma \epsilon)$	Pluggable: Stochastic Resonance, Quantum-inspired gates
Plasticity Socket (Learning)	Updates weights	$\Delta W = \eta \cdot (R \cdot \epsilon)$	Pluggable: STDP, Metaplasticity

Precision-Weighted Predictive Brake:

$$\frac{d\theta}{dt} = \omega \cdot \exp(-\beta \epsilon^2) + K \sum_i w_i \sin(\psi_i - \theta)$$

Extensibility and Future-Proofing:

The socket-based design ensures that new computational modules, learning rules, or gating mechanisms can be seamlessly integrated. Each socket acts as a **pluggable interface**, allowing researchers or hardware designers to introduce novel predictive gates, plasticity schemes, or error-computation strategies **without affecting the stability or core functionality** of existing neurons. This modularity makes P-TUC 2.0 inherently **future-proof**, adaptable to evolving computational paradigms, and compatible with both neuromorphic hardware and conventional architectures.

5. Phase 5 — Hierarchical & Asynchronous Validation

Bidirectional Flow & Error Blending:

$$\epsilon_{\text{total}} = \alpha \cdot \epsilon_{\text{local}} + (1 - \alpha) \cdot \epsilon_{\text{feedback}}$$

- Layer N clocks are constrained by Layer $N - 1$.
- **Fast Path:** Familiar inputs $\rightarrow$ Structural Intuition
- **Slow Path:** Novel inputs $\rightarrow$ Structural Deliberation

Emergent Sparsity:

- Silent neurons consume minimal energy (ϵ high $\rightarrow$ Ground State)
- Only resonant neurons use power

6. Phase 6 — Failure Mode Analysis: The Paradox Response

- **Contradiction:** $\epsilon \rightarrow \max \Rightarrow$ clock stops $\rightarrow$ permanent silence
- **Pure Noise:** $R < \tau_{\text{crit}} \Rightarrow$ Ground State

Input Type	Standard AI Response	P-TUC 2.0 Response	Physical State
Clear Truth	Immediate Output	High-Speed Tick	Full Resonance
Ambiguity	Statistical Guess	Delayed Tick	Evidence Accumulation
Contradiction	Hallucinated Logic	Permanent Silence	Clock Freeze
Pure Noise	Pattern Seeking	Ground State	Sub-critical Coherence

7. Phase 7 — Formal Analysis: Convergence to Ground State

Potential Function:

$$V(\theta) = -\omega_{\text{eff}} \cdot \theta - \sum_i K_i \cos(\psi_i - \theta)$$

Driving Force at Limit:

$$\omega_{\text{eff}} = \omega_0 \cdot \exp(-\beta \epsilon^2) \approx 0 (\epsilon \rightarrow \infty)$$

Incoherence Attractor:

$$\frac{d\theta}{dt} \approx K \sum_i \sin(\psi_i - \theta) \rightarrow 0$$

Realization Barrier:

$$\int \frac{d\theta}{dt} d\tau > \tau_{\text{crit}} \Rightarrow y = 0$$

8. Self-Critique & Mitigation Summary

- **Complexity:** Reduced via asynchronous processing
 - **Hardware:** Tunable T allows microsecond scaling
 - **Scalability:** Local learning avoids backprop bottlenecks
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9. Comparative Context: P-TUC vs Other Architectures

Proposal	Learning Mechanism	Output Validation	Scalability	Strengths	Limitations
Standard Transformers	Global Backprop	Forced Softmax	High	Mature, widely used	Hallucinations, energy-heavy
Forward-Forward (Hinton)	Local “Goodness”	Positive vs Negative Data	Medium	Layer-wise local learning	No temporal coherence; soft outputs
Equilibrium Prop (Bengio)	Energy Minimization	Settles on min energy	Low	Physics-inspired, removes backprop	Two-phase, slow convergence
Active Inference (Friston)	Surprise Minimization	Rejects high-entropy inputs	Low	Brain-like, principled uncertainty	Bayesian-heavy, difficult to scale
P-TUC (Proposed)	Local Phase-Lock + Predictive Error	Structural Silence (Ground State)	High	Energy-efficient, neuromorphic-native, predictable output control	Requires tuning of T ; untested at scale

10. Conclusion

- P-TUC 2.0 enforces structural constraints promoting reliable outputs
 - Coherence-driven responses emerge naturally
 - Dishonesty is **structurally constrained**, not externally imposed
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11. P-TUC 2.0 — Equations Reference Table

Phase	Equation	Variables	Notes
1	$y(t) = f(Wx)$ if stable; 0 otherwise	$y(t), W, x, T$	Ground State prevents output from unstable input
2	$\epsilon(t) = S_{ff} - P_{fb} $	Local error	Internal calculation

Phase	Equation	Variables	Notes
2	$\frac{d\theta}{dt} = \omega_0 \cdot \text{sech}(\gamma\epsilon) + K\sum \sin(\psi_i - \theta)$	θ, γ	Error-gated PLL
2	$y(t) = f(Wx)$ if $\int \epsilon^2 d\tau < \tau_{\text{threshold}}$; 0 otherwise	$\tau_{\text{threshold}}$	Output only if error low
3	$\Delta W = \eta \cdot (R \cdot \epsilon)$	R, η	Local update on successful tick
4	$\frac{d\theta}{dt} = \omega \cdot \exp(-\beta\epsilon^2) + K\sum w_i \sin(\psi_i - \theta)$	β, w_i	Precision-weighted brake
5	$\epsilon_{\text{total}} = \alpha \cdot \epsilon_{\text{local}} + (1 - \alpha) \cdot \epsilon_{\text{feedback}}$	α	Upper layer constrained by lower layer
6	$\epsilon_{\text{paradox}} = \lim (S_{\text{ff}} - P_{\text{fb}}) \approx \max$	Max error	Clock stops for contradictions
6	$y(t) = 0$ if $\int R d\tau < \tau_{\text{crit}}$	R	Prevents noise output
General	T	Integration window	Tunable 500 ms $\rightarrow$ microseconds
Socket 1	Comparator	$\epsilon = S_{\text{ff}} - P_{\text{fb}} $	Pluggable Error Logic
Socket 2	Gating	Brake $= \text{sech}(\gamma\epsilon)$	Predictive Brake
Socket 3	Plasticity	ΔW $= \eta \cdot (R \cdot \epsilon)$	Learning Rules

12. Limitations and Future Perspectives

While P-TUC 2.0 provides a rigorous structural solution to the problem of "Forced Realization," several engineering and theoretical challenges remain as targets for future research.

A. The "Cascade of Silence" (Deadlock Risk)

In a purely inhibitory system, there is a risk that high levels of input noise could cause a "Cascade of Silence," where a network becomes overly cautious and fails to produce any output.

Mitigation: Future iterations will explore **Stochastic Resonance**, where a controlled "noise floor" allows the Inference Clock to overcome minor predictive gaps, ensuring the system remains responsive to faint but valid signals.

B. Cumulative Latency in Deep Hierarchies

Because each layer requires a temporal Integration Window (T) to validate its predecessor, a deep network could suffer from significant **"Time-to-First-Tick" latency**.

Future Work: Research into **Dynamic T-Scaling**, where the integration window shrinks as a function of historical confidence (the "Fast Path" for familiar patterns), reducing latency for routine processing while maintaining **Structural Deliberation** for novel inputs.

C. Consistency vs. Objective Truth

P-TUC 2.0 ensures **Structural Honesty** (preventing the model from inventing data internally). However, it cannot inherently correct for **"Consistent Misinformation"** if the training data itself is logically coherent but factually false.

Future Work: Integrating **External Verifiers** into the Comparator Socket to cross-reference internal resonance with verified knowledge graphs or real-world sensory data.

D. Hardware-Software Divergence

Current GPU architectures are optimized for synchronous, dense matrix operations. P-TUC's asynchronous, differential-equation-based nature is better suited for **Neuromorphic Hardware** (e.g., Spiking Neural Networks).

Future Work: Developing **P-TUC Kernels** specifically for hybrid architectures that can simulate asynchronous phase-locking efficiently on parallel hardware.

E. Scaling to Billion-Parameter Models

While the $O(1)$ Local Learning rule is mathematically efficient, its convergence properties at the scale of **Large Language Models (LLMs)** remain untested.

Future Work: Large-scale simulations to determine how **Hierarchical Error Blending** handles the extreme dimensionality of transformer-level parameter spaces.

Summary:

Despite these limitations, **P-TUC 2.0's core structural honesty and modular extensibility remain intact**, ensuring that its key strengths — coherent output, energy efficiency, and predictive gating — continue to provide a reliable foundation for future AI research and neuromorphic applications.